

Exercise-1 (Setting Up JUnit in Eclipse)

Step 1: Create a Java Project

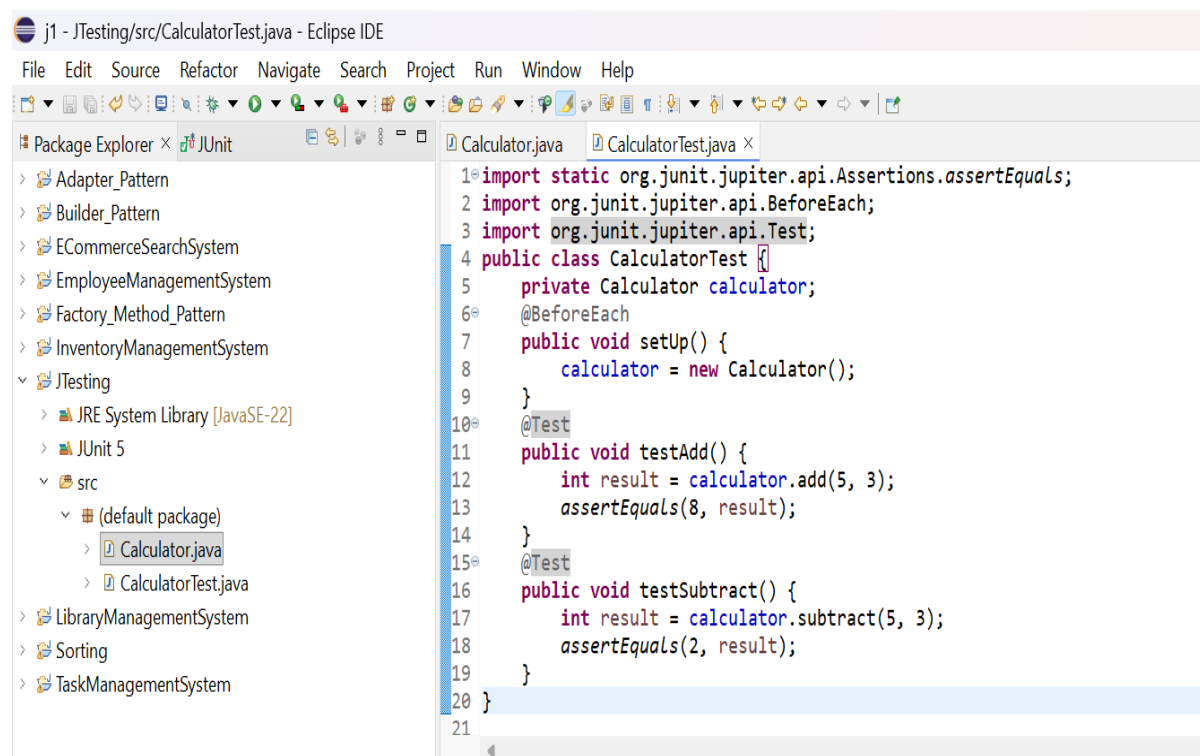
1. Open Eclipse.
2. Click **File** → **New** → **Java Project**.
3. Enter Project Name: JTesting
4. Click **Finish**.

Step 2: Add JUnit 5 Library

1. Right-click the project.
2. Select **Build Path** → **Add Libraries**.
3. Choose **JUnit**.
4. Click **Next**.
5. Select **JUnit 5**.
6. Click **Finish**.

Step 3: Create Calculator Class

1. Right-click **src**.
2. Select **New** → **Class**.
3. Class Name: Calculator
4. Click **Finish**.



The screenshot shows the Eclipse IDE interface. The top bar indicates the current file is 'j1 - JTesting/src/CalculatorTest.java - Eclipse IDE'. The menu bar includes File, Edit, Source, Refactor, Navigate, Search, Project, Run, Window, and Help. The Package Explorer on the left shows the project structure: Adapter_Pattern, Builder_Pattern, ECommerceSearchSystem, EmployeeManagementSystem, Factory_Method_Pattern, InventoryManagementSystem, JTesting (expanded), JRE System Library [JavaSE-22], JUnit 5, src (expanded), (default package) (expanded), Calculator.java, and CalculatorTest.java. The main editor area shows the code for CalculatorTest.java:

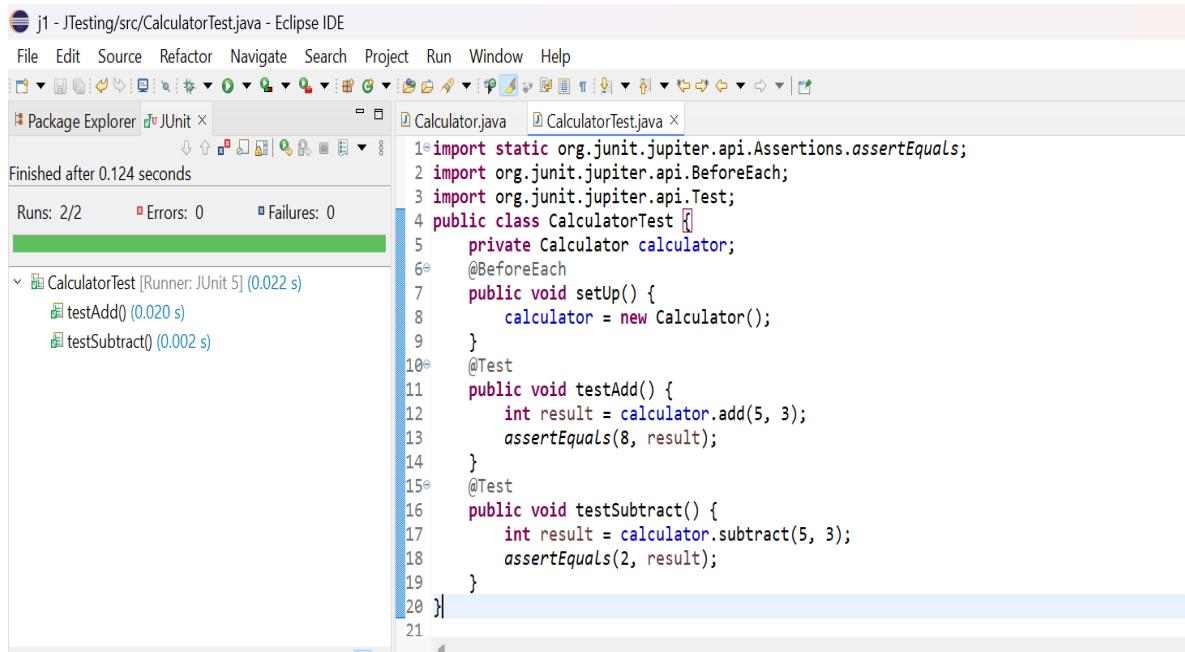
```
1 import static org.junit.jupiter.api.Assertions.assertEquals;
2 import org.junit.jupiter.api.BeforeEach;
3 import org.junit.jupiter.api.Test;
4 public class CalculatorTest {
5     private Calculator calculator;
6     @BeforeEach
7     public void setUp() {
8         calculator = new Calculator();
9     }
10    @Test
11    public void testAdd() {
12        int result = calculator.add(5, 3);
13        assertEquals(8, result);
14    }
15    @Test
16    public void testSubtract() {
17        int result = calculator.subtract(5, 3);
18        assertEquals(2, result);
19    }
20 }
21
```

Step 4: Create Test Class

1. Right-click **src**.
2. Select **New** → **Class**.
3. Class Name: CalculatorTest
4. Click **Finish**.

Step 5: Run JUnit Test

1. Right-click **CalculatorTest.java**.
2. Click **Run As** → **JUnit Test**.



Expected Output

In the JUnit tab:

Runs: 2/2

Errors: 0

Failures: 0

A **green bar** indicates all tests passed successfully.